

BANKHAUL · REQUIREMENTS

BankHaul requirements

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	BKH-REQ-001	Version	0.2
Status	Draft	Date	2026-10-03
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/bankhaul		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-03	Amish Chadha	TRL 3; targets made measurable, R10 (weak link) and R11 (working distance) added, status from BKH-CAL-001
0.1	2026-09-30	Amish Chadha	Initial scaffold

BankHaul requirements

Measurable requirements for the first prototype station. Each status comes from the calculation note BKH-CAL-001 (docs/04-calcs/01-sizing.md); "on paper" means calculated, not yet tested. R10 and R11 were added at the TRL 2 review (BKH-DDR-001, items 3 and 4).

Table 1. Requirements

ID	Requirement	Target	Verification (TRL 4 or later)	Status at TRL 3
R1	Set and haul a gillnet with nobody entering the water	10 of 10 set and haul cycles completed from the bank	Field trial with partner fishers	Met on paper: once the far stake is set from a boat, no step needs anyone in the water
R2	Working span	At least 30 m (98 ft) between pulleys; stretch goal 50 m (164 ft)	Measured at the trial site	Met on paper: 30 m design case; rope bought for 50 m
R3	Hand pull force	Peak pull under 250 N (56 lbf) to haul a 30 m net with catch	Spring scale or load cell during trials	Met on paper: about 200 N (estimate)
R4	Bank anchor holding	Holds at least 3 kN (675 lbf) in soft wet soil without visible movement	Pull-out test on a CalRig-style rig	Met on paper: screw anchors at factor 2.1 on capacity in soil of 10 kPa shear strength
R5	Offshore stake or float holding	Holds at least 1.5 kN (337 lbf) horizontal load in soft mud	Pull-out tests in a representative bed	Met in soft mud (factor 2.5); at risk in very soft mud (factor 1.2)
R6	Cycle time	Full haul and reset under 10 minutes for a 30 m net	Timed trials	Met on paper: about 6 minutes (estimate, without clearing fish)
R7	Durability	One fishing season (6 months) of daily use without rope or pulley failure	Season-long pilot with inspection log	Met on paper (estimate): rope factor 5.7 after sun damage
R8	Cost	Parts under USD 80 per station	Costed bill of materials from local prices	Not met: USD 382 per station
R9	Level change	Bank anchor relocatable by two people in under 30 minutes as the shoreline moves	Timed relocation	Met on paper: about 22 minutes (estimate)
R10	Weak link	The net's bridle releases between 500 and 700 N, so a net seized by an animal or snag cannot load the loop, stake or fisher beyond that	Pull three samples of each cord batch to release	Met on paper; cord chosen by test when parts are bought
R11	Working distance	The fisher's working spot is at least 5 m (16 ft) from the water's edge at the day's level	Measured at the trial site	Met by layout: 5.0 m

■ Assumptions

- Fishers can shift from wading-set nets to loop-set nets without losing much catch.
- Most attack hotspots have banks where a screw anchor can hold (soft wet soil of about 10 kPa undrained shear strength or firmer).
- A boat or low-water window exists to place the far stake once per season.

- Nets can be fitted with a bridle at each end to clip to the loop and the bank post.
- Design case: a 30 m by 2 m gillnet, 10 kg of catch and 5 kg of weed; water 1.0 m deep at the far stake.

Safety: R10 and R11 are safety requirements. They limit what a seized net can do to the fisher and keep the fisher back from the water. Neither makes it safe to stand at the water's edge.